

Operating Systems — Course Syllabus

Unit 1: Process Management

A process is a program in execution, consisting of the program code, current activity, and associated resources. The operating system is responsible for process creation, scheduling, and termination. Process scheduling algorithms include First-Come-First-Served (FCFS), Shortest Job First (SJF), Round Robin, and Priority Scheduling, each balancing throughput, turnaround time, and fairness differently.

Unit 2: CPU Scheduling and Threads

A thread is the smallest unit of CPU execution within a process, allowing concurrent execution paths that share the same memory space. Multithreading improves responsiveness and resource utilization compared to single-threaded processes. Context switching is the mechanism by which the CPU switches from executing one process or thread to another, saving and restoring state as needed.

Unit 3: Deadlocks

A deadlock occurs when a set of processes are blocked because each process is holding a resource and waiting for another resource held by another process. The four necessary conditions for deadlock are mutual exclusion, hold and wait, no preemption, and circular wait. Deadlock can be handled through prevention, avoidance using algorithms like the Banker's Algorithm, detection, or recovery.

Unit 4: Memory Management

Memory management techniques include paging, which divides memory into fixed-size blocks called pages and frames, and segmentation, which divides memory based on logical program divisions. Virtual memory allows a process to execute without being entirely loaded into physical memory, using techniques like demand paging and page replacement algorithms such as FIFO, LRU, and Optimal.